

Lab 03 - Audit

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Topic: [ADM_BD_03.ppt](#)

Lab: [Lab03_auditoria_trigger.txt](#)

Execution instructions

Each item of the experiment must be carried out and its result documented in the report.

Lab objective

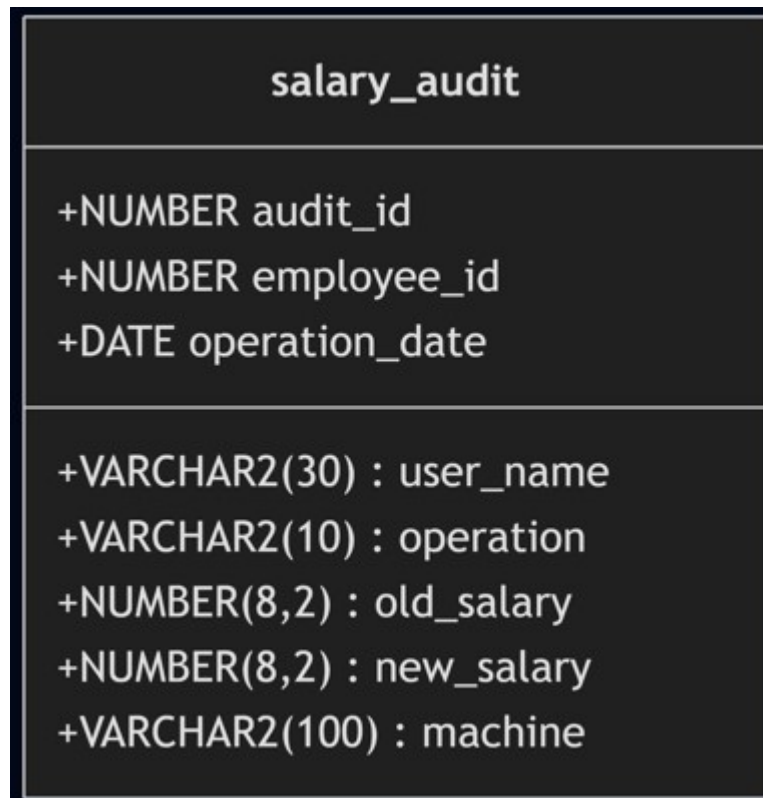
The goal of this lab is to implement a database audit mechanism using a trigger in the Oracle HR schema. The main objective is to detect and document suspicious salary changes made by users who exploit the system by increasing their own salary just before payday (5th of each month), and then reverting it back afterward.

As the DBA, create the necessary tools to:

- Log salary changes between the 1st and 5th of each month.
- Identify abnormal behavior that might indicate policy violations or fraud.

Diagram

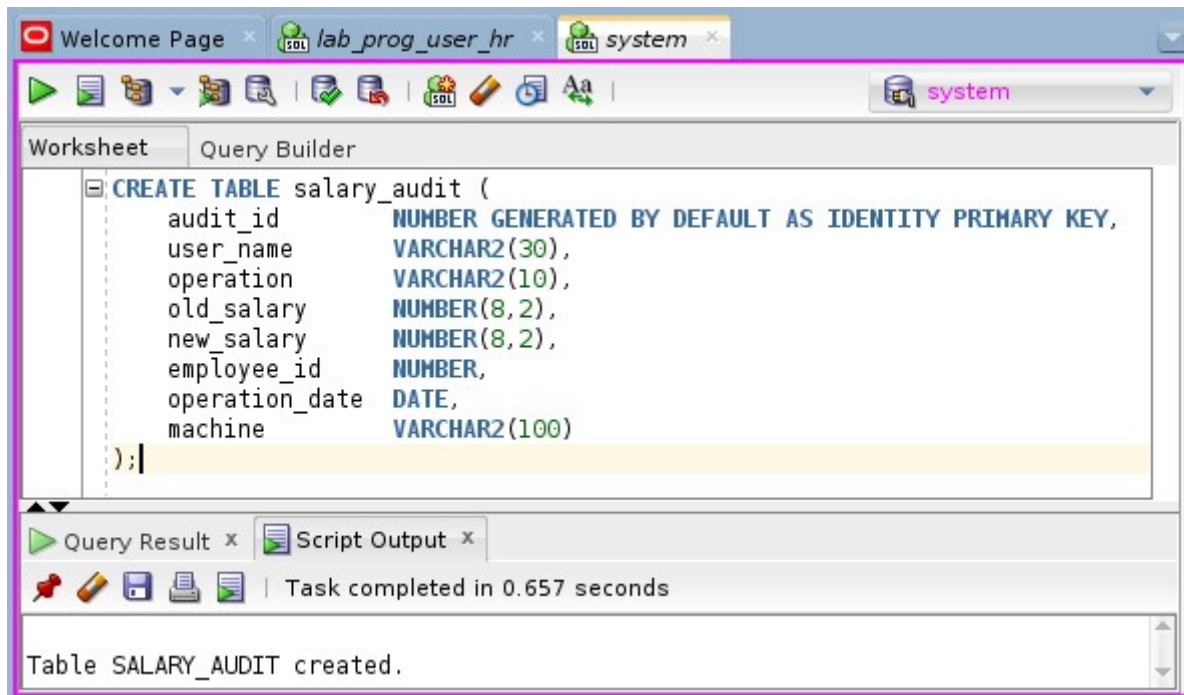
```
classDiagram
class salary_audit {
+NUMBER audit_id
+VARCHAR2(30) user_name
+VARCHAR2(10) operation
+NUMBER(8,2) old_salary
+NUMBER(8,2) new_salary
+NUMBER employee_id
+DATE operation_date
+VARCHAR2(100) machine
}
```



Img 01 - salary_audit TABLE Diagram

DDL

```
CREATE TABLE salary_audit (  
  audit_id      NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  user_name     VARCHAR2(30),  
  operation     VARCHAR2(10),  
  old_salary    NUMBER(8,2),  
  new_salary    NUMBER(8,2),  
  employee_id   NUMBER,  
  operation_date DATE,  
  machine       VARCHAR2(100)  
);
```



Img 02 - Executes an CREATE TABLE query

Audit Trigger

```
CREATE OR REPLACE TRIGGER trg_salary_audit
BEFORE INSERT OR UPDATE OR DELETE ON employees
FOR EACH ROW
DECLARE
  v_user      VARCHAR2(30);
  v_machine   VARCHAR2(100);
  v_now       DATE := SYSDATE;
BEGIN
  v_user := SYS_CONTEXT('USERENV', 'SESSION_USER');
  v_machine := SYS_CONTEXT('USERENV', 'HOST');

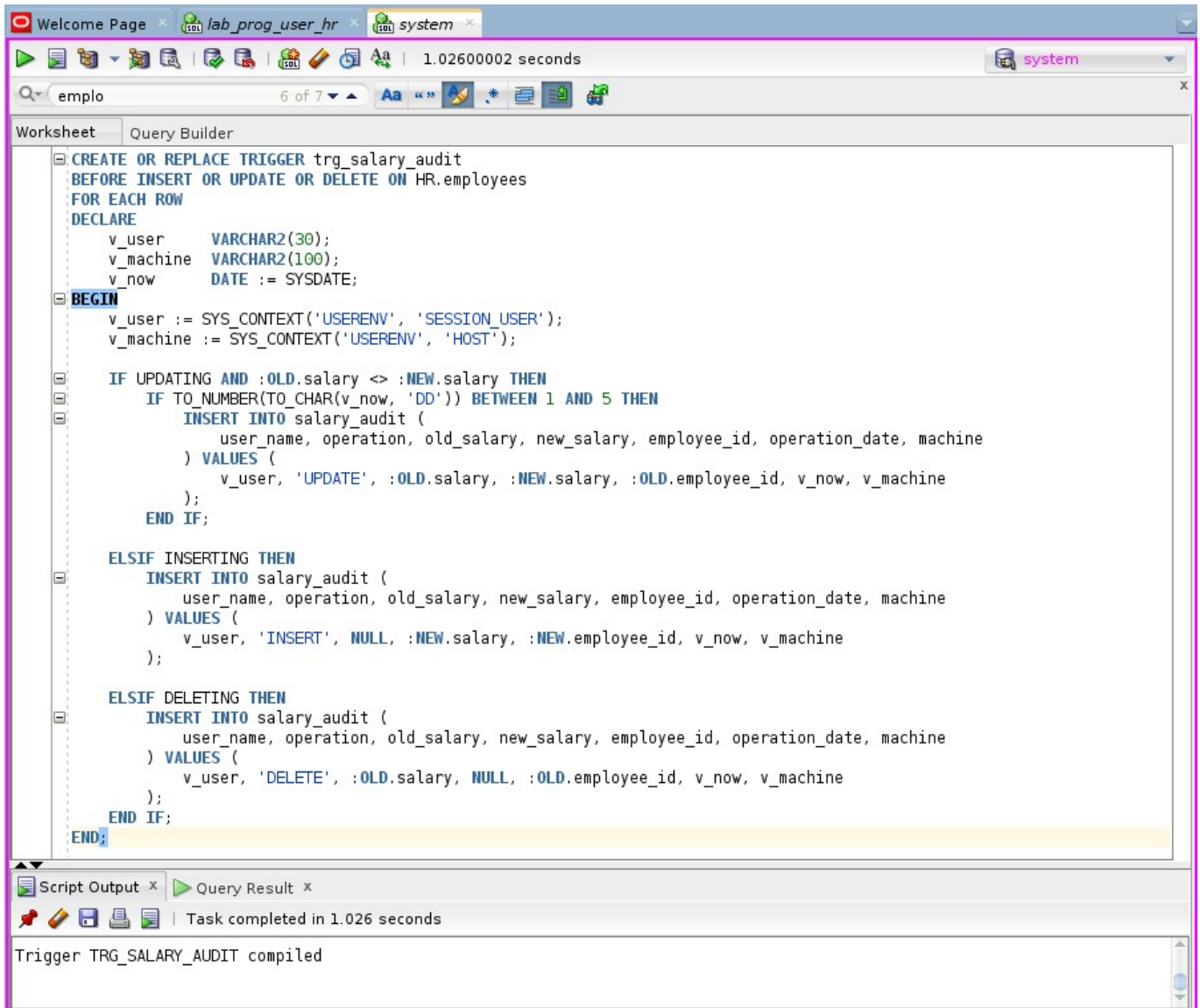
  IF UPDATING AND :OLD.salary <> :NEW.salary THEN
    IF TO_NUMBER(TO_CHAR(v_now, 'DD')) BETWEEN 1 AND 5 THEN
      INSERT INTO salary_audit (
        user_name, operation, old_salary, new_salary, employee_id,
        operation_date, machine
      ) VALUES (
        v_user, 'UPDATE', :OLD.salary, :NEW.salary, :OLD.employee_id,
        v_now, v_machine
      );
    END IF;

  ELSIF INSERTING THEN
    INSERT INTO salary_audit (
      user_name, operation, old_salary, new_salary, employee_id,
      operation_date, machine
    ) VALUES (
      v_user, 'INSERT', NULL, :NEW.salary, :NEW.employee_id, v_now,
      v_machine
    );
```

```

    ELSIF DELETING THEN
        INSERT INTO salary_audit (
            user_name, operation, old_salary, new_salary, employee_id,
            operation_date, machine
        ) VALUES (
            v_user, 'DELETE', :OLD.salary, NULL, :OLD.employee_id, v_now,
            v_machine
        );
    END IF;
END;

```



Img 03 - Executes the TRIGGER creation query